Assessment for Jr. Data Scientist Position

Pathao Limited

Submitted By,

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Part 1 Exploratory Data Analysis :

a) Given the provided dataset can you provide insights on how request volume and Completion Rate (CR = total_completed_trips/total_requests) changes throughout time? Do you think request volume is an accurate measure of demand?

b) Driver profiling is a critical component for reliability analysis+optimization. Given the dataset, can you identify metrics that can measure driver performance/preferences? Can you then compute these metrics for each driver to create individual performance profiles? (an example of a performance metric would be Assignment to completion ratio (ACR) = total_completed_trips/total_assigned_requests).

Can you plot the distributions of the metrics you identified?

Solution:

a) Here to gather insights on how request volume and Completion Rate (CR = total_completed_trips /total_requests) changes throughout time –
   - Introduced a function named **Get_total_request()** that counts the total number of request assign at each unique time(Hour).
   - Introduced a function named **Get_Completed_riding_request()** that counts the total number of request assign which were completed by the driver at each unique time(Hour).
   - Then by using total_completed_trips and total_requests which were calculated by using the following functions we calculated Completion Rate (Completion_Rate_CR).
   - Then I introduced a line-plot and a count-plot to observe insight on how request volume changes throughout time at each unique time(Hour).
   - Then I introduced a line-plot and a count-plot to observe insight on how Completion Rate (CR = total_completed_trips/total_requests) changes throughout time at each unique time(Hour).

Findings : From Line-plot and Count plot we observe that-

- Request volume changes throughout time (at each unique time(Hour) where it is high around 9 and 10th hour and low during 1 to 4th hour in a day.
- Completion Rate changes throughout time (at each unique time(Hour) where it is high around 12,16 and 17th hour and low during 1 and 2nd hour in a day.

Do you think request volume is an accurate measure of demand?

Yes, I think request volume which means how many request comes in a particular time is definitely an accurate measure of demand. An only by observing the riding request we can have insight about which particular time in a day a passenger needs a ride in general. According to which the organization can take all the necessary to keep the system available.

Source Code :

https://github.com/urmisen/Assignments/blob/main/Pathao_assignment_main_Part_1_(A)_request_volume_and__CR.ipynb

b) Driver profiling is a critical component for reliability analysis +optimization. Given the dataset, can you identify metrics that can measure driver performance/preferences? Can you then compute these metrics for each driver to create individual performance profiles? (an example of a performance metric would be Assignment to completion ratio (ACR) = total_completed_trips/total_assigned_requests).
Can you plot the distributions of the metrics you identified?

Solution:

Here to introduce the Driver profiling system and to measure driver performance/preferences we

introduced new metrics Assignment to completion ratio (ACR)= total_completed_trips /total_assigned_requests for which we-

- Introduced a function named **Get_assign_request()** that counts the total number of request assign to each driver.
- Introduced a function named **Completed_riding_request()** that counts the total number of request assign which were completed for each individual driver.
- Then by using total_completed_trips and total_assigned_requests which were calculated by using the following functions we calculated Assignment to completion ratio (ACR) = total_completed_trips/total_assigned_requests).
- Introduced a function named **Ride_distance_Op()** which generates the average ride distance for which each driver completed the trip.
- Introduced a function named **Ride_duration_Op()** which generates the average ride duration for which each driver completed the trip.
- Introduced a function named **Ride_duration_Op()** which generates the average ride duration for which each driver completed the trip.
- Introduced a function named **Avg_Rating_Op()** which generates the average rating for which each driver completed the trip.
- Taking all these generated metrics we created a CSV file to create Driver Profiling Dataset.
- Now we ploted the distributions of the metrics that is identified using distplot() and pairplot().
- Finally I ploted a heatmap() to observe the correlation between the feature metrics.

Findings: From the distribution we can observe that,

- By observing the multivariate analysis from the heapmap() we see that the distribution of the metrics are not normal and there existslots of outliers in the dataset.
- The correlation between Average ride distance and number of total trip for a particular driver id is negative, which means if the ride distance is small then the possibility of monthly total trip for a driver is high.
- The correlation between Average ride duration and number of total trip for a particular driver id is negative, which means if the ride duration is low then the possibility of monthly total trip for a driver is high.
- Assignment to completion ratio and Total trip is positive. For obvious reason if the completion ratio is high then the total trip count end of the month will also be high.

Source Code:

https://github.com/urmisen/Assignments/blob/main/Pathao_assignment_Part_1_(B)_Driver_profiling.ipynb

Part 2 Predictive Modeling

Oftentimes it is important to estimate a driver's future performance when calculating supply allocation requirements.

Based on the one week's worth of data provided, can you design a model that predicts a driver's end of month completed trips? The completed trip dataset can be used as labels for supervised models.

Do not fret too much about model accuracy; we are more interested in the relevancy of the features you engineer for the model.

Solution:

Here I have used the generated Driver Profiling Dataset and completed trip dataset and introduced a linear regression model to predict a driver's end of month completed trips based on the one week's worth of data.

Again to do feature scaling I have used MinMaxScaler() which is a skearn preprocessing function that transform features by scaling each feature to a range.

I have used Ensemble method to combine multiple models to produce improved result. Ensemble methos usually produces more accurate solution than a single model would do.

Source Code : https://github.com/urmisen/Assignments

Part 3 PoC System Design

Let's assume Pathao wishes to improve the Assignment to completion ratio (ACR). They wish to create a "smart" dispatch system that assigns a request to the driver most likely to accept it.

Imagine drivers A, B, C are potential candidates to assign to request X. The dispatch system would choose driver "B" where driver "B" has the highest probability of accepting the request.

Using the driver profiling data you prepared in part 1b and any additional insights you obtained, can you design a Proof of Concept algorithm that sorts and assigns the most likely driver to a given request?

Do you feel additional data outside of what was provided could make the algorithm more robust? If yes, mention the data points and how you would use them.

I have also used GridSearchCV() technique to do hyper-parameter tuning.

Solution:

Here I have used driver profiling dataset that I have prepared in part 1b to design a Proof of Concept algorithm that sorts and assigns the most likely driver to a given request. For which I have followed the following steps :

1. Initially I will take the riding distance of the request assignment.
2. Find the difference between the riding distance and the Average riding distance for each driver id, and then we took the absolute value of the difference.

   Diff= abs(Riding Distance-Average Riding Distance for each driver id)
3. keep track of the index values,

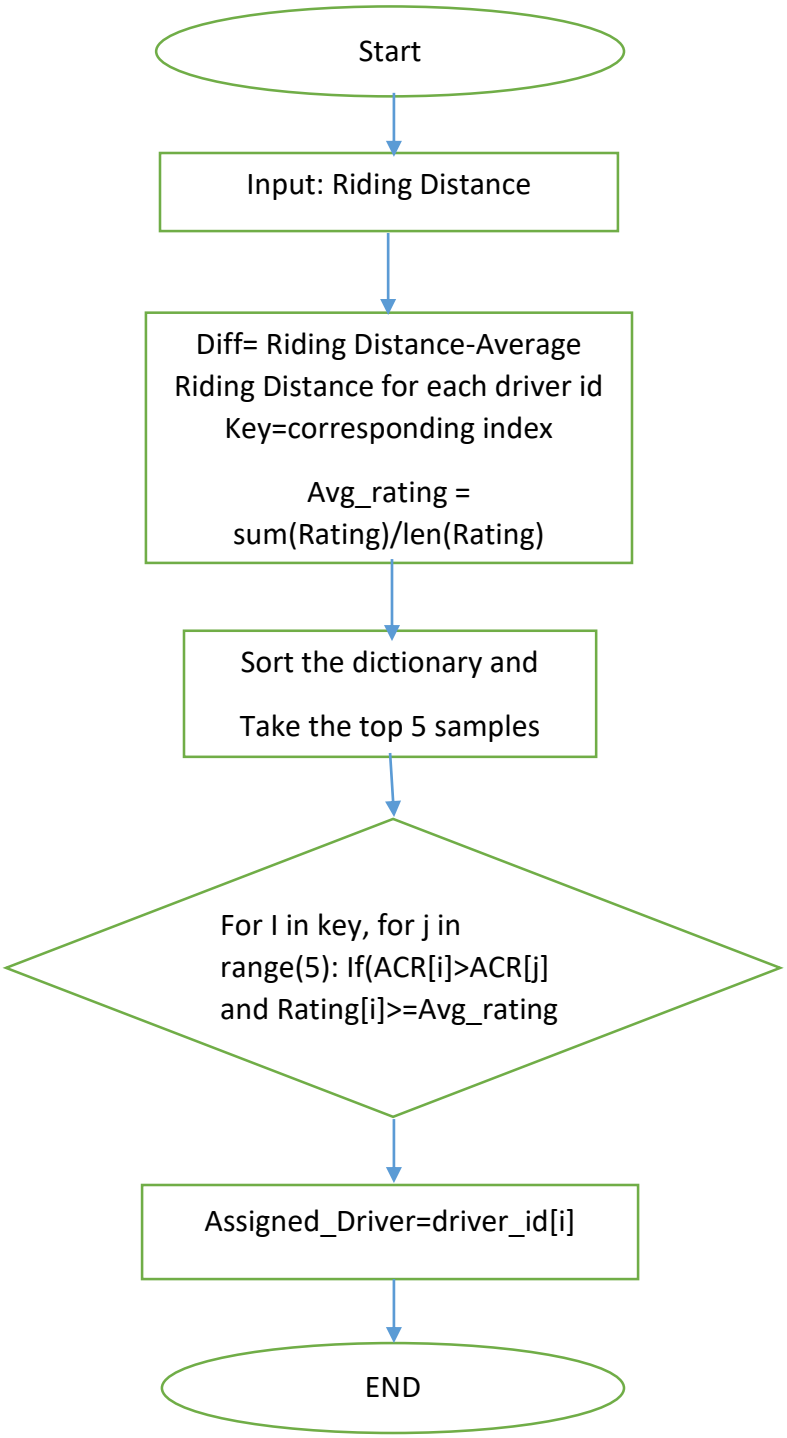
   Key=corresponding index
4. Create a dictionary using key and diff data.

   dict(key:diff)
5. Sort the dictionary from lower to high and keeps the track of the keys.
6. Find the maximum "assignment to completion ratio" keep in mind about the minimum riding difference.
7. If the Rating of the corresponding driver is also high then only the riding request will assign to that particular driver id.

Source Code:

https://github.com/urmisen/Assignments/blob/main/Pathao_assignment_part_3_Proof_of_the_Concept_algorithm_.ipynb

Flowchart:


Start
Input: Riding Distance
Diff= Riding Distance-Average Riding Distance for each driver id
Key=corresponding index
Avg_rating = sum(Rating)/len(Rating)
Sort the dictionary and
Take the top 5 samples
For I in key, for j in range(5): If(ACR[i]>ACR[j] and Rating[i]>=Avg_rating
Assigned_Driver=driver_id[i]
END